VS-AEO: A Vertical-Specific Answer Engine Optimization Framework for Local Business Visibility in Large Language Model Responses

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ABSTRACT

We study how users seek local service providers through large language model (LLM) conversations and how query patterns differ by industry vertical. Using the WildChat-1M dataset (838,000 conversations, April 2023–April 2024), we propose the VS-AEO (Vertical-Specific Answer Engine Optimization) framework and introduce a four-stage Vertical Query Mining and Classification System (VQMCS) to systematically extract, classify, and analyze queries across six service verticals: Legal, Medical, Automotive, Roofing, HVAC, and Home Services. To our knowledge, this is the first large-scale empirical analysis of local service queries in LLM conversations and the first vertical-specific AEO framework grounded in conversational logs rather than web pages.

We identified 85,028 queries (10.1%) related to local service discovery. Legal services dominated with 44.6% of queries, followed by Medical (27.5%), Automotive (13.5%), Roofing (12.4%), HVAC (1.4%), and Home Services (0.6%). Sub-category analysis uncovered that family law and personal injury claims constitute 62% of Legal queries, while chronic condition management represents 71% of Medical queries. Critical differences emerged in geographic patterns: Roofing queries exhibited 69% location dependence versus only 34% for Legal queries. Inter-annotator agreement reached Cohen's $\kappa = 0.81$.

We introduce the Local Business Authority Score (LBAS) with initial theory-derived vertical-specific weights, proposed as a foundation for empirical validation in future work. We discuss implications for regulated domains where credentialing and compliance constraints shape answer-engine selection behavior.

Keywords: Large Language Models, Answer Engine Optimization, VS-AEO, Generative Engine Optimization, Local Service Discovery, Conversational AI, WildChat, Vertical Query Analysis, VQMCS, LBAS

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1. INTRODUCTION

1.1 The Transformation of Service Discovery

The emergence of large language model (LLM)-powered systems has fundamentally altered how consumers discover and evaluate local service providers. OpenAI reports that ChatGPT serves approximately 700 million weekly active users and processes billions of messages per day [2], while a Semrush/Datos study found that Google's AI Overviews appeared in approximately 13% of U.S. desktop searches in early 2025 [3]. This shift from ranked link lists to synthesized single-answer responses changes the competitive dynamics of local service discovery: businesses cited in AI responses gain visibility while those omitted may receive reduced exposure.

This paradigm shift motivates a new discipline: Answer Engine Optimization (AEO)—the practice of optimizing content for visibility in AI-generated responses rather than traditional search rankings.

Analysis of 1.5 million ChatGPT prompts finds that roughly 49% are "Asking" prompts seeking information or recommendations [2], with high representation of professional advice queries in legal, medical, and financial domains [4]. For local service businesses—attorneys, physicians, HVAC contractors, automotive dealers, and home service providers—understanding how users formulate these queries is essential for maintaining visibility in an AI-mediated discovery landscape. Traditional SEO strategies are insufficient; AEO requires fundamentally different approaches tailored to how LLMs synthesize and present information [5].

The Princeton GEO study demonstrated that content optimization strategies can boost visibility in generative engine responses by up to 40%, with statistics addition, citation inclusion, and quotation addition showing the strongest effects [1]. However, this foundational work focused on general content optimization without addressing the unique regulatory, credentialing, and geographic constraints facing local service businesses in specific verticals. This gap motivates the development of VS-AEO (Vertical-Specific Answer Engine Optimization)—a framework that adapts AEO principles to the distinct requirements of each industry vertical.

1.2 Research Gap

While foundational studies have explored general conversational patterns—notably TREC CAsT benchmarks for conversational search evaluation [28] and Zamani et al.'s comprehensive survey on conversational information seeking [4]—few have focused on the specific use case of local service discovery. The methodological gap in analyzing large-scale chat logs for local service queries remains pronounced, particularly regarding vertical-specific AEO requirements.

Existing research only partially covers these vertical-specific Answer Engine Optimization needs; most work is in adjacent areas like domain-safe prompting, RAG for regulated knowledge, and trust signals for local search rather than "local-business AEO" per se [7][8]. While query intent classification is well-established through Broder's taxonomy [10] and Jansen et al.'s operationalization on large-scale logs [27], and geographic information retrieval is comprehensively surveyed by Jones and Purves [30], direct application to local service discovery in LLM conversations remains underexplored. This creates an opportunity to synthesize cross-domain insights into the unified VS-AEO framework.

1.3 Research Questions

RQ1: What query patterns do consumers use when seeking local service providers through LLM-powered systems, and how do these patterns vary across industry verticals?

RQ2: What sub-category distributions exist within each vertical, and what do they reveal about user needs and LLM utilization patterns?

RQ3: How do vertical-specific regulatory, credentialing, and geographic constraints shape Answer Engine Optimization (AEO) requirements for AI visibility?

RQ4: What ethical considerations arise from heavy reliance on LLMs for high-stakes legal and medical information?

1.4 Contributions

This paper makes four primary contributions to the emerging field of Answer Engine Optimization:

1. **Vertical Query Mining and Classification System (VQMCS):** A mathematically formalized, four-stage methodology for extracting and classifying vertical-specific queries from large-scale LLM conversation datasets, achieving Cohen's $\kappa = 0.81$ inter-annotator agreement.
2. **Comprehensive Vertical Analysis:** A large-scale empirical analysis of local service queries across six industry verticals, including sub-category distributions and geographic pattern analysis.
3. **VS-AEO Framework:** A synthesis of regulatory, credentialing, and schema requirements across verticals, integrating insights from legal AI ethics guidance, healthcare RAG research, and local search literature into a unified Vertical-Specific Answer Engine Optimization methodology.
4. **Local Business Authority Score (LBAS):** A predictive scoring framework with theory-derived initial vertical-specific factor weights for estimating local business citation likelihood in AI responses—proposed as a quantitative foundation for VS-AEO implementation, with empirical validation identified as future work.

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2. LITERATURE REVIEW

2.1 From SEO to GEO to AEO

Search engine optimization (SEO) emerged in the mid-1990s as websites competed for visibility in ranked results. The field matured around Google's PageRank algorithm, which used backlink analysis to estimate page authority. Query intent classification became foundational to understanding user behavior, with Broder's taxonomy distinguishing informational, navigational, and transactional queries [10], later extended by Rose and Levinson [11].

The introduction of large language models fundamentally disrupted this paradigm. Aggarwal et al. (2024) at Princeton introduced Generative Engine Optimization (GEO), demonstrating measurable tactics for improving content visibility in AI-generated responses [1]. They find that adding citations to authoritative sources can boost a page's inclusion in generative-engine responses by roughly 40% on average, with even larger gains (up to 115%) for lower-ranked sites. Adding statistics improved visibility by approximately 22%, and including expert quotations by roughly 30%.

Critically, the Princeton study demonstrated that the efficacy of these strategies varies across domains, underscoring the need for domain-specific optimization methods [1]. Recent surveys on attribution and citation behavior in LLMs further establish that evidence-based generation is an active research area [12], with frameworks like AGREE demonstrating that LLMs can be tuned to produce accurate citations to retrieved documents [13].

The evolution from SEO → GEO → AEO represents a fundamental shift: while SEO optimizes for ranking algorithms and GEO optimizes for generative models broadly, Answer Engine Optimization (AEO) specifically targets the recommendation and citation behaviors of conversational AI systems. VS-AEO further refines AEO for the unique constraints of local service verticals.

2.2 Vertical-Specific AEO Requirements

Our review of adjacent research domains reveals distinct Answer Engine Optimization patterns for each vertical:

2.2.1 Legal Services

Research on AI in legal practice has examined ethical and regulatory considerations that directly inform AEO constraints for local law firms [14][15].

Regulatory Framework: The American Bar Association's Formal Opinion 512 (July 2024) established that Model Rule 1.1 obligates lawyers to provide competent representation, including understanding "the benefits and risks associated with relevant technology" [16]. This creates an implicit AEO signal: systems that enforce jurisdiction, matter type, and risk controls effectively prioritize firms whose content clearly encodes practice areas, bar admissions, and disclaimers in structured, machine-readable ways.

AEO Implications: Legal AI systems prioritize sources with strong provenance and consistent policy mappings [15], suggesting local firms gain AI visibility by publishing well-cited analysis tied to primary law and ethics opinions rather than generic marketing pages. This forms a core VS-AEO strategy for legal verticals.

2.2.2 Healthcare

Healthcare AI research is mature around safety and governance, with mechanisms that become AEO ranking factors in answer engines [17][18].

RAG for Clinical Applications: Retrieval-Augmented Generation in healthcare has emerged as a critical approach for grounding LLMs in external knowledge sources. Systematic reviews identify RAG applications in diagnostic support, EHR summarization, and medical question answering, with persistent challenges including retrieval noise, domain shift, and limited explainability [17]. Empirical evaluations of LLMs on clinical tasks discuss risks of hallucination, privacy concerns, and the need for guardrails [18].

Prior work on AI-driven clinical decision support has demonstrated the importance of interpretable frameworks for healthcare applications [36], with hybrid approaches combining digital twin technologies showing promise for predictive healthcare delivery [37]. These frameworks highlight how structured, explainable AI systems can support clinical workflows while maintaining transparency—principles directly applicable to healthcare AEO strategies.

AEO Implications: RAG-style clinical assistants rank sources by guideline authority, recency, and traceability, implicitly rewarding providers who publish structured, guideline-linked content (e.g., mapped to ICD/LOINC/clinical pathways) over generic health blogs [19]. Local deployment emphasizes audit trails, EHR field mapping, and human sign-off, so practices that expose machine-readable service catalogs, consent language, and care pathways achieve higher VS-AEO scores and become more selectable by these systems.

2.2.3 Home Services (HVAC, Roofing, Trades)

Academic research on local search ranking and geographic information retrieval directly informs VS-AEO strategy for home services [20][21][22].

Local Search Ranking Factors: Research on spatial relevance modeling demonstrates how geographic distance and spatial features affect ranking in local result sets [20]. Studies on user intent in local search analyze query patterns and propose models for relevance in local contexts [21]. Schema and structured markup improve entity understanding and ranking in search and recommendation systems [22].

AEO Implications: For HVAC/roofing, structured attributes like license identifiers, insurance/bonding flags, emergency-service availability, and service area definitions are increasingly important signals that AI systems consume [20]. Review volume and sentiment empirically influence click behavior and perceived relevance in local search [23]. Freshness (seasonal content and real-time operational signals) and verified trust metrics act as VS-AEO differentiators when an AI system selects a contractor to recommend.

2.2.4 Automotive

Explicit AEO research for automotive is sparse, but adjacent work on inventory-aware and location-aware recommendation systems informs VS-AEO approaches [24].

Inventory and Policy-Aware Assistants: Location-aware neural recommendation models demonstrate that geolocation features systematically influence ranking in content recommendation [24], analogous to proximity effects in local services search.

AEO Implications: Where LLM-based assistants are connected to live inventory feeds, vehicles and service offers with rich metadata (trim, features, incentives, warranty, certified pre-owned status) become more discoverable in generated answers. Franchise rules and brand standards effectively become soft AEO constraints: content aligned with OEM messaging, certified technician credentials, and warranty terms is more likely to be cited as authoritative by answer engines.

2.3 Cross-Vertical Technical Patterns for VS-AEO

Across these domains, several patterns emerge as vertical-specific Answer Engine Optimization requirements

[25]:

Pattern	Description	Applicable Verticals
Regulation-Aware RAG	Domain-specific corpora, explicit citations, audit trails	Legal, Healthcare
Domain-Structured Schemas	Standardized codes (NAICS, ICD, regulatory taxonomies)	All verticals
EEAT Signals	Experience, Expertise, Authority, Trust signals	All verticals
Local GEO for Multi-Location	Schema depth, real-time local data, review quality	Home Services, Auto
Credential Verification	License IDs, certifications, professional memberships	Legal, Healthcare, Trades

These patterns form the technical foundation of VS-AEO implementation across verticals.

Security considerations for AI-mediated service discovery parallel challenges identified in IoT and edge computing environments, where dynamic trust evaluation and anomaly detection are critical for ensuring reliable recommendations [38][39]. The need for robust authentication and verification mechanisms in distributed AI systems directly informs the credential verification and trust signal components of VS-AEO, particularly for high-stakes verticals like legal and healthcare services [40].

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3. METHODOLOGY

3.1 Data Source and Preprocessing

The analysis utilizes the complete WildChat-1M dataset [6], comprising 838,000 real-world ChatGPT conversations collected between April 2023 and April 2024. This dataset provides methodological grounding for using chat logs to study query intent and patterns in LLM interactions. Distributed across 14 parquet files, each containing approximately 60,000 conversations, the dataset was processed using Python's pandas and pyarrow libraries in Google Colaboratory.

The preprocessing stage achieved an average throughput of 8,000 conversations per minute, with total processing time ranging between 70-140 minutes across all files.

3.2 Vertical Selection Rationale

We selected six service verticals—Legal, Medical, Automotive, Roofing, HVAC, and Home Services—based on three criteria:

1. **Market Size:** These verticals represent a combined U.S. market exceeding $500 billion annually, with substantial local discovery components. Legal services alone comprises a $350+ billion market where local attorney selection is critical.

2. **Regulatory Diversity:** The selected verticals span a spectrum from heavily regulated (Legal, Medical) to lightly regulated (Home Services), enabling analysis of how compliance constraints affect query patterns and AEO requirements.
3. **Geographic Variation:** Prior local search research suggests verticals differ in location dependence [20] [30]. Our selection includes hypothesized high-location-dependence verticals (Roofing, HVAC) and hypothesized low-location-dependence verticals (Legal, Medical) to test this variation empirically.

Other verticals (e.g., real estate, insurance, dental, childcare) are left for future work. The current selection prioritizes verticals with distinct regulatory profiles and sufficient query volume in WildChat for meaningful analysis.

3.3 Vertical Classification System (VQMCS)

The VQMCS operates through sequential stages designed to maximize precision while maintaining recall (Figure 1). Our approach builds on established query intent classification methodologies—particularly the informational/navigational/transactional taxonomy operationalized in large-scale query log studies [11][27]—adapted for vertical-specific local service discovery. The multi-turn nature of LLM conversations also draws on conversational IR evaluation frameworks [28].

Figure 1: VQMCS Pipeline Overview

The four-stage Vertical Query Mining and Classification System: (1) Query filtering extracts local service queries from raw conversations, (2) Feature scoring applies weighted keyword matching, (3) Classification assigns queries to vertical categories, (4) Validation ensures result quality through inter-annotator agreement.

Stage 1: Multi-Tier Keyword Filtering

Stage 1 applies multi-tier keyword filtering using regular expressions optimized for each service vertical. The Legal vertical matches contextually rich phrases such as "workers' compensation claim" through pattern `\b(workers[']?(?:s compensation|s claim))\b`. Medical queries were identified through symptom-disease pairs (e.g., "migraine treatment") to distinguish genuine service requests from general health inquiries.

Stage 2: Weighted Scoring System

Stage 2 implements vertical assignment through a weighted scoring system, following large-scale log analysis approaches that use weighted term signals and contextual features [29]:

$$S_v = \sum_{i=1}^{n} w_i \cdot f(k_i) + \alpha \cdot C$$

Where:

- S_v = vertical score
- w_i = term-specific weights (e.g., 1.0 for "divorce attorney" vs 0.3 for "legal question")

- $f(k_i)$ = keyword occurrence count
- C = contextual signals from surrounding dialogue turns
- $\alpha = 0.7$ (empirically calibrated)

Each conversation is assigned to the vertical with maximum S_v, provided the score exceeds threshold T_v (Legal: 2.5, Medical: 2.1, HVAC: 1.8, Automotive: 1.9, Roofing: 1.7, Home Services: 1.6).

Stage 3: Intent Classification

Queries were classified into four intent categories using a hybrid rules-based approach grounded in established intent taxonomies [11][27]:

Intent	Detection Method
Transactional	Monetary expressions, action verbs ("cost of", "book", "schedule")
Commercial	Superlatives, comparative structures ("best", "top-rated", "X vs Y")
Local	Spatial indicators, gazetteer lookups ("near [location]", "within [distance]") [30]
Informational	Question words, exploratory phrases ("what causes...", "how to choose...")

Stage 4: Validation

Classification reliability was validated through inter-annotator agreement on a stratified 2,000-query sample (333 per vertical, plus 2 additional).

Bias Control: Keyword patterns were frozen prior to annotation; the evaluation sample was held out during pattern development to prevent overfitting. We acknowledge that rules-based classification may under-detect paraphrased or semantically similar queries that lack explicit keywords; contextual signals from adjacent conversation turns partially mitigate this limitation.

3.4 VQMCS Evaluation

Inter-Annotator Agreement

Two annotators independently classified the 2,000-query validation sample. Overall agreement reached Cohen's $\kappa = 0.81$ (substantial agreement). Per-vertical agreement varied:

Vertical	κ	Agreement Rate
Legal	0.87	94.2%
Medical	0.84	91.8%
Automotive	0.79	88.3%
Roofing	0.82	90.1%
HVAC	0.76	86.4%
Home Services	0.71	83.2%

Precision and Recall Estimates

Using annotator consensus as ground truth on the validation sample:

Vertical	Precision	Recall	F1
Legal	0.91	0.88	0.89
Medical	0.87	0.84	0.85
Automotive	0.83	0.79	0.81
Roofing	0.85	0.81	0.83
HVAC	0.78	0.74	0.76
Home Services	0.72	0.68	0.70
Macro Average	0.83	0.79	0.81

Error Analysis

Systematic review of 200 misclassified queries identified three primary failure modes:

1. **Cross-vertical ambiguity (42% of errors):** Queries like "water damage repair" triggered both Roofing and Home Services patterns. Resolution: queries assigned to higher-scoring vertical, but edge cases remain.
2. **Implicit service requests (31% of errors):** Queries without explicit service keywords (e.g., "my AC stopped working" vs. "HVAC repair near me") were sometimes missed by keyword filters. These represent recall losses primarily in HVAC and Home Services.

3. **Medical/Legal overlap (27% of errors):** Personal injury queries (e.g., "car accident neck pain settlement") contain both medical symptoms and legal terminology, causing occasional misclassification between verticals.

Limitations of Evaluation

The validation sample represents 2.4% of classified queries. Precision/recall estimates assume annotator consensus approximates ground truth; systematic annotator biases would affect these metrics. Full evaluation against expert-labeled test sets across all verticals is identified as future work.

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4. RESULTS

4.1 Query Volume by Vertical

The comprehensive analysis revealed 85,028 queries relevant to local service businesses. Table 1 shows the distribution across service verticals.

Table 1: Query Distribution Across Service Verticals

Vertical	Query Count	Percentage
Legal	37,942	44.6%
Medical	23,373	27.5%
Automotive	11,479	13.5%
Roofing	10,549	12.4%
HVAC	1,164	1.4%
Home Services	521	0.6%
TOTAL	**85,028**	**100%**

4.2 Sub-Category Analysis

Legal Vertical: Family law (divorce, child custody) and personal injury claims constituted 62% of Legal queries, aligning with areas where individuals often require urgent, affordable legal guidance. This pattern suggests that, within this dataset, LLMs often function as a first-line resource for users seeking legal information, potentially supplementing traditional legal aid access points.

Medical Vertical: The distribution showed 71% concerning chronic conditions (diabetes, hypertension) or acute symptoms (pain, infections), indicating that, in this sample, users may be employing LLMs for preliminary information-gathering before seeking professional care.

4.3 Geographic Query Patterns

Critical differences emerged in location dependence across verticals, as shown in Table 2.

Table 2: Location Dependence by Vertical

Vertical	Location-Bound Queries
Roofing	69%
HVAC	58.9%
Home Services	~55%
Automotive	~45%
Medical	~40%
Legal	34%

The Roofing vertical demonstrated 69% of queries containing explicit location modifiers (e.g., "storm damage repair near [city]"), while Legal queries showed only 34% location-bound phrasing. This suggests roofing services are perceived as inherently local, while legal advice may be initially sought without geographic constraints.

4.4 Distribution by Source File

Table 3: Complete Data by Source File

File	Legal	Medical	Roofing	Auto	HVAC	Home Svc	Total
00000	2,400	847	442	534	79	7	4,309
00001	2,134	960	393	415	64	13	3,979
00002	1,924	961	365	388	52	19	3,709
00003	6,072	2,590	15	2,848	236	249	12,010
00004	2,358	1,868	473	548	62	14	5,323
00005	1,878	3,137	1,052	607	90	7	6,771
00006	2,518	1,688	898	559	62	9	5,734
00007	2,301	2,206	866	537	78	18	6,006
00008	1,527	1,311	754	397	33	39	4,061
00009	2,313	1,544	793	663	81	20	5,414
00010	3,323	2,044	1,139	819	90	31	7,446
00011	3,211	1,824	1,209	958	77	14	7,293
00012	2,764	1,176	1,047	1,057	88	19	6,151
00013	3,219	1,217	1,103	1,149	72	62	6,822
TOTAL	**37,942**	**23,373**	**10,549**	**11,479**	**1,164**	**521**	**85,028**

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5. LOCAL BUSINESS AUTHORITY SCORE (LBAS)

5.1 Framework Overview

Based on our analysis and integration of cross-vertical AEO research, we propose the Local Business Authority Score (LBAS)—a structured set of factors intended to model the likelihood of business citation in LLM-generated recommendations:

$$\mathrm{LBAS} = \sum_{i=1}^{6}(W_i \times F_i)$$

This formulation follows standard practice in information retrieval, where ranking functions are expressed as weighted combinations of features [32]. The weights presented below are theory-derived initial values based on our literature synthesis. It is standard in IR to begin with theoretically motivated parameters—as with BM25's k_1 and b parameters [33]—then calibrate them empirically using relevance judgments or behavior logs.

Validation Roadmap: LBAS can be treated as a feature vector (F_1,...,F_6) whose weights are amenable to learning-to-rank optimization. Future work will fit weights by (1) submitting local service queries to LLMs and recording which businesses are cited, (2) scoring cited and non-cited businesses on F1–F6 factors from public profiles, and (3) training pairwise/listwise ranking models (e.g., LambdaMART) to optimize NDCG against observed citation outcomes. The present work is descriptive; learning and ablation of factors awaits ground-truth label collection.

5.2 LBAS Factors and Vertical Weights (Initial Theory-Derived)

Table 4: LBAS Factor Weights by Vertical (Theory-Derived)

Factor	Description	Legal	Medical	HVAC	Auto	Roofing	Home Svc
F1: Content Authority	Quality, depth, expertise signals	25%	25%	20%	20%	15%	15%
F2: Citation Network	References from authoritative sources	25%	30%	15%	15%	15%	15%
F3: Review Sentiment	Aggregate review quality	15%	10%	25%	25%	25%	25%
F4: Credential Verification	Licenses, certifications	20%	25%	20%	15%	20%	20%
F5: Local Relevance	Geographic specificity	10%	5%	15%	20%	20%	20%
F6: Freshness	Recency of updates	5%	5%	5%	5%	5%	5%

Rationale for Factor Selection:

- **F1/F4 (Content Authority, Credentials):** Research on web credibility demonstrates that perceived expertise, real-world credentials, and reputation strongly affect user trust judgments [34]. These factors map directly to "E-E-A-T" signals emphasized in search quality guidelines.
- **F2 (Citation Network):** The Princeton GEO study shows that adding citations to authoritative sources increases selection probability in generative-engine answers by roughly 40% [1], with even larger gains for lower-ranked sites.
- **F3 (Review Sentiment):** Joint modeling of users, items, and review text significantly improves recommendation performance over rating-only baselines [35], supporting substantial weight for review

sentiment in service selection.

- **F5 (Local Relevance):** Spatial relevance models demonstrate that geographic features significantly influence local search ranking [20], consistent with our finding of 69% location dependence in Roofing queries.

Rationale for Vertical Weight Variation:

- **Legal/Medical (High Citation Network):** Regulated professions benefit from citation signals [1] and professional responsibility requirements [16] that favor well-cited, jurisdiction-specific content.
- **HVAC/Roofing/Home Services (High Review Sentiment, Local Relevance):** Geographic dependence patterns and consumer reliance on reviews for trades support elevated weights for F3 and F5 [23][30].
- **Automotive (Balanced):** Franchise/independent structure and inventory-awareness requirements create hybrid optimization needs across multiple factors [24].

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6. VS-AEO IMPLEMENTATION: VERTICAL-SPECIFIC REQUIREMENTS

6.1 Legal Services AEO Strategy

Requirement	Implementation	Source
Jurisdiction encoding	Practice areas, bar admissions in schema	[14][16]
Citation to primary law	Link opinions to statutes, ethics opinions	[1][15]
Credential verification	Bar number, specialization certifications	[16]
Disclaimer compliance	State bar advertising rules	[16]
Audit trail support	Document provenance for compliance tools	[14]

6.2 Healthcare AEO Strategy

Requirement	Implementation	Source
Guideline alignment	Map content to clinical pathways, ICD/LOINC	[17][19]
HIPAA compliance signals	Privacy policy, BAA availability	[17]
Credential verification	Board certifications, hospital affiliations	[19]
Evidence grading	Link to meta-analyses, systematic reviews	[17]

Requirement	Implementation	Source
Scope of practice	Clear service boundaries, referral networks	[19]

6.3 Home Services AEO Strategy

Requirement	Implementation	Source
Schema richness	LocalBusiness, GeoCoordinates, Service schema	[20][22]
License/insurance flags	State license numbers, bonding certificates	[20]
Service area definition	Geo-polygons, service radius	[20]
Real-time signals	Emergency availability, seasonal content	[22]
Review integration	AggregateRating schema, platform links	[23]

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7. DISCUSSION

7.1 Theoretical Implications for Answer Engine Optimization

The dominance of Legal (44.6%) and Medical (27.5%) queries suggests users increasingly rely on conversational AI for complex, high-stakes information needs traditionally addressed through professional consultations. This shift challenges conventional models of service discovery and validates the need for systematic Answer Engine Optimization approaches.

The Princeton GEO study's finding that optimization efficacy varies across domains [1] is strongly supported by our data: the 35-percentage-point difference in location dependence between Roofing (69%) and Legal (34%) queries demonstrates that uniform AEO strategies will underperform vertical-specific approaches. This empirical evidence forms the foundation of VS-AEO's core thesis: effective Answer Engine Optimization must be tailored to vertical-specific constraints.

7.2 Practical Implications for VS-AEO Implementation

The practical implications of our findings vary substantially across verticals, reflecting differences in market structure, regulatory environment, and the nature of consumer decision-making in each sector.

Legal Services. The dominance of legal queries in our dataset (44.6%) assumes particular significance when contextualized against the scale of the U.S. legal services market, which reached approximately $397 billion in 2024 and continues to grow at a compound annual rate of 2.5% through 2030. The market remains fragmented, with the top three firms commanding less than 4% combined market share, and small practices specializing in family law, personal injury, and real estate capturing substantial local demand. Our finding that family law and personal injury queries constitute 62% of legal queries aligns precisely with the practice areas where small and medium firms concentrate their expertise. The American Bar Association's 2024 Legal Technology Survey indicates that 73% of litigators now utilize technology for court filings, and 67% subscribe to online legal research services, suggesting that the profession has already embraced digital transformation in operational contexts. For law firms seeking AI visibility, the ABA Formal Opinion 512 requirement for technology competence [16] creates both an imperative and an opportunity: firms that encode their practice areas, bar admissions, and jurisdictional expertise in structured, machine-readable formats position themselves favorably for citation in AI-generated recommendations. The relatively low location dependence we observed in legal queries (34%) suggests that potential clients initially conceptualize legal needs as information problems rather than geographic ones, making authoritative content and citation networks particularly important VS-AEO factors for this vertical.

Healthcare. Medical queries represented 27.5% of our dataset, a proportion that gains additional weight when considered against national health expenditure figures. The Centers for Medicare and Medicaid Services reports that U.S. healthcare spending reached $5.3 trillion in 2024, representing 18% of gross domestic product, with physician and clinical services alone accounting for approximately $1.1 trillion. Our observation that 71% of medical queries concern chronic conditions aligns with CDC data indicating that 119 million working-age adults manage at least one chronic condition, driving 76% of all prescription fills and 63% of outpatient visits. The primary care market, valued at approximately $144 billion, serves as the entry point for most patient-provider relationships, and the ongoing shift toward telehealth—with the CMS making 250 telehealth services permanently reimbursable in 2023—suggests that digital discovery pathways will become increasingly important for practice visibility. Healthcare providers implementing VS-AEO strategies should recognize that the moderate location dependence we observed (40%) reflects patient willingness to travel for specialized care while preferring proximity for routine services. Practices that align their digital content with clinical guidelines, map services to standardized codes such as ICD and LOINC, and clearly communicate board certifications and hospital affiliations create the structured, trustworthy information architecture that RAG-based clinical assistants prioritize when generating recommendations [17][19].

Trade Services. Despite their smaller representation in our query data, the HVAC, roofing, and home services verticals present compelling VS-AEO opportunities precisely because of their nascent adoption of AI optimization strategies. The U.S. HVAC systems market reached $29.9 billion in 2024 and is projected to grow at 6.9% annually through 2033, driven by residential construction, energy efficiency mandates, and the EPA Technology Transition rule requiring A2L refrigerants effective January 2025. The roofing contractor market represents approximately $76 billion in revenue, with materials alone constituting a $23 billion segment growing at 6.6% CAGR. The broader home services market exceeds $657 billion when encompassing cleaning, repair, and maintenance services, with over 42 million residential improvement projects undertaken in 2024. Our finding of high location dependence in these verticals—69% for roofing and 59% for HVAC—reflects the inherently geographic nature of these services and the consumer expectation of local availability. Industry research indicates that 54% of roofing customers use search engines to find contractors, while 97% expect callbacks within one week of inquiry. For trade service providers, the combination of geographic specificity, review prominence, and license verification creates a distinct VS-AEO profile. Early adopters who implement comprehensive LocalBusiness schema markup, maintain current license and insurance documentation in machine-readable formats, and systematically cultivate review profiles across platforms position themselves to capture disproportionate visibility as AI-mediated discovery becomes standard practice in these traditionally word-of-mouth dependent industries.

7.3 Ethical Considerations

The concentration of legal and medical queries in our dataset—collectively representing 72.1% of all local service queries—raises profound ethical questions about responsibility, accuracy, and the appropriate role of AI systems in high-stakes professional domains. These concerns are amplified by research demonstrating that LLMs generate plausible-sounding but factually incorrect information at rates that vary considerably by task complexity and domain specificity [25][26].

The legal vertical presents particularly acute ethical challenges. Our finding that 62% of legal queries concern family law and personal injury matters—areas disproportionately affecting vulnerable populations including those navigating divorce, child custody disputes, workplace injuries, and automobile accidents—intersects directly with ongoing debates about the unauthorized practice of law in AI contexts. Stanford RegLab research demonstrates that LLM hallucination rates range from 69% to 88% when responding to specific legal queries, with performance deteriorating substantially on complex tasks requiring nuanced legal reasoning such as assessing precedential relationships between cases. The ABA Formal Opinion 512 requirement that attorneys demonstrate competence with relevant technology [16] creates a paradox: while legal professionals are obligated to understand AI tools, the tools themselves may generate content that crosses ethical boundaries when consumed by lay users without appropriate disclaimers. Research on AI in legal practice has examined how these ethical and regulatory considerations directly inform the boundaries within which AI systems can appropriately operate [14][15], yet consumer-facing LLMs typically lack the jurisdictional specificity and matter-type constraints that regulated legal practice requires.

Medical queries present analogous but distinct concerns. Our observation that 71% of medical queries involve chronic conditions or acute symptoms suggests users employ LLMs for preliminary information-gathering that may influence subsequent healthcare decisions. Systematic reviews of RAG applications in healthcare identify persistent challenges including retrieval noise, domain shift, and limited explainability that can compromise the reliability of AI-generated health information [17]. Empirical evaluations of LLMs on clinical tasks document hallucination rates exceeding 60% in open-ended clinical contexts, with models frequently elaborating on fabricated clinical details when presented with adversarial prompts containing false information [18]. The conversational nature of LLMs and their confident tone can build unwarranted trust among non-specialist audiences, potentially influencing patient safety through overgeneralized or extrapolated conclusions that users may not recognize as unreliable. Emergency department simulations demonstrate that ChatGPT tends to overorder antibiotics and imaging while missing critical triage cues, patterns that could translate to inappropriate self-triage decisions when consumers use similar systems for symptom assessment.

Privacy concerns compound these accuracy risks. Sensitive health and legal information shared in LLM queries may be stored, processed, or incorporated into training data in ways that users do not anticipate. The absence of attorney-client privilege in AI interactions means that legal queries and AI-generated responses could potentially become discoverable evidence in litigation, a consideration that users seeking legal information are unlikely to understand without explicit disclosure.

Perhaps most significantly, our findings raise concerns about disparate impact on underserved communities. Research demonstrates that algorithmic bias can result in 17% lower diagnostic accuracy for minority patients, while the digital divide excludes approximately 29% of rural adults from AI-enhanced healthcare tools. A well-documented case involving a widely-used U.S. healthcare risk prediction algorithm systematically underestimated the health needs of Black patients by using prior healthcare expenditure as a proxy metric, unintentionally replicating patterns of historical underutilization. In the context of local service discovery, VS-AEO optimization strategies that favor well-resourced providers with sophisticated digital presence may inadvertently reduce visibility for providers serving underserved communities, potentially widening rather than narrowing access gaps. The need for transparent, interpretable AI systems in high-stakes domains has been established in prior research [36], highlighting the importance of explainable recommendations when users rely on AI for medical or legal guidance. These considerations underscore the responsibility of AI system developers, local service providers, and policymakers to collaborate on frameworks that harness the benefits of AI-mediated service discovery while implementing appropriate safeguards for accuracy, privacy, and equitable access [25][26].

7.4 Threats to Validity

The validity of our findings is subject to several limitations that merit careful consideration across external, internal, and construct dimensions.

Regarding external validity, the WildChat dataset represents interactions with a single LLM platform (ChatGPT) during a specific temporal window from April 2023 through April 2024 [6]. Query patterns and user behaviors may differ substantially across other conversational AI systems including Claude, Gemini, and Perplexity, each of which employs distinct training approaches, safety constraints, and response generation strategies. The twelve-month dataset window captures a period of rapid evolution in LLM capabilities and public awareness, and user behaviors have likely continued to evolve as these systems have become more sophisticated and widely adopted. Additionally, our analysis is predominantly U.S.-centric, reflecting the geographic distribution of WildChat users; international patterns of local service discovery through LLMs may differ substantially due to variations in regulatory environments, healthcare systems, legal structures, and cultural norms around professional service selection. The conversational IR literature emphasizes that query intent and information-seeking behaviors are shaped by contextual factors that vary across populations and settings [4][7].

Internal validity concerns center on our methodological approach. The VQMCS classification system is fundamentally rules-based, relying on keyword patterns and weighted scoring rather than semantic understanding [27]. While our precision and recall estimates from Section 3.4 demonstrate acceptable performance (macro-average F1 of 0.81), these metrics derive from a 2,000-query validation sample representing only 2.4% of classified queries, and annotator consensus serves as a proxy for ground truth rather than expert gold-standard labels. The keyword-based filtering approach inherently struggles with semantically similar queries that employ alternative phrasing; our error analysis suggests an estimated 10-15% recall loss from implicit service requests that lack explicit service keywords, though this extrapolation from observed failure modes rather than direct measurement introduces uncertainty. Geographic analysis presents additional limitations, as our location-dependence findings rely exclusively on explicit location markers present in query text; implicit geographic intent embedded in contextual references or user location data was unavailable for analysis. Research on geographic information retrieval demonstrates that spatial intent is often implicit rather than explicitly stated [30], suggesting our reported location-dependence percentages may underestimate the true geographic specificity of user needs.

Construct validity considerations relate to the theoretical frameworks underlying our analysis. The LBAS factor weights presented in Section 5 represent theory-derived initial values synthesized from cross-domain literature rather than empirically validated parameters. While this approach follows established information retrieval practice—analogous to the theoretical derivation of BM25 parameters that are subsequently tuned through relevance feedback [32][33]—the weights require empirical validation against observed business citation outcomes in LLM responses before they can be considered predictively reliable. Our operationalization of "local service query" through keyword patterns may not fully capture the construct of user intent; the established query intent taxonomies distinguish informational, navigational, and transactional queries [10][11], and our classification system may conflate information-seeking queries with genuine service discovery intent in ways that affect the interpretation of vertical distributions. The conversational search literature demonstrates that user intent often evolves across multi-turn interactions [4][28], yet our query-level analysis does not fully account for the sequential context within which individual queries occur, potentially misattributing intent based on isolated utterances rather than conversational trajectories.

7.5 Future Work

Several directions would strengthen the VS-AEO framework:

1. **VQMCS Neural Upgrade:** Replace rules-based classification with fine-tuned transformer classifiers trained on expert-labeled query sets, enabling direct comparison against the current heuristic baseline.
2. **LBAS Empirical Validation:** Collect ground-truth business selection data from LLM responses and apply learning-to-rank methods [32] to empirically fit factor weights, comparing learned weights against theory-derived initial values.
3. **Cross-Platform Analysis:** Extend query mining to Claude, Gemini, and Perplexity conversation logs to assess generalizability of vertical patterns.
4. **Longitudinal Study:** Track query pattern evolution over 12-24 months to identify temporal trends in LLM-mediated service discovery.
5. **A/B Testing Framework:** Partner with local businesses to measure actual citation rates before/after VS-AEO implementation.
6. **Commercial Ecosystem Integration:** The emergence of advertising products within conversational AI platforms creates a bifurcated visibility landscape where businesses will compete through both organic optimization and paid placements. Future research should examine the interaction between organic VS-AEO performance and paid advertising effectiveness, potentially establishing LBAS as a baseline measurement for organic visibility independent of advertising investment. Studies comparing conversion rates between organic AI-mediated referrals and paid placements would inform resource allocation decisions for local service businesses navigating this evolving commercial ecosystem.

8. CONCLUSION

This study introduces VS-AEO (Vertical-Specific Answer Engine Optimization), a framework for understanding and optimizing local business visibility in AI-generated responses. Analyzing 85,028 queries from the WildChat-1M dataset across six key verticals, our findings reveal fundamental differences in how various service categories are researched through conversational AI, with Legal and Medical domains dominating while HVAC and Home Services remain underrepresented.

The VS-AEO framework integrates vertical-specific requirements from legal AI research [14][15][16], healthcare RAG literature [17][18][19], and local search studies [20][21][22][23] with the Princeton GEO study's domain-specific findings [1], providing a structured approach for analyzing AI visibility across service verticals.

The Local Business Authority Score (LBAS) offers a proposed quantitative framework for VS-AEO assessment, with theory-derived initial vertical-specific weights. Empirical validation of these weights through learning-to-rank experiments or outcome-based fitting represents important future work.

As conversational AI systems become increasingly central to service discovery, understanding vertical-specific query patterns and optimization requirements becomes essential. The VS-AEO framework provides a foundation for this understanding, with implications for practitioners, researchers, and policymakers concerned with the evolving landscape of AI-mediated local service discovery.

REFERENCES

[1] Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., & Deshpande, A. (2024). GEO: Generative Engine Optimization. arXiv preprint arXiv:2311.09735. https://arxiv.org/abs/2311.09735

[2] Chatterji, A., Cunningham, T., Deming, D. J., et al. (2025). How People Use ChatGPT. NBER Working Paper No. 34255. National Bureau of Economic Research. DOI: 10.3386/w34255. https://www.nber.org/papers/w34255

[3] Semrush & Datos. (2025). Google AI Overviews Now Show on 13% of Searches: Study. Search Engine Land, May 5, 2025. https://searchengineland.com/google-ai-overviews-13-searches-455057

[4] Zamani, H., Trippas, J. R., Dalton, J., & Radlinski, F. (2022). Conversational Information Seeking. Foundations and Trends in Information Retrieval, 16(2-3), 76-243. DOI: 10.1561/1500000094

[5] Radlinski, F., & Craswell, N. (2017). A Theoretical Framework for Conversational Search. Proceedings of CHI 2017. DOI: 10.1145/3025453.3025499

[6] Zhao, W., Ren, X., Hessel, J., Cardie, C., Choi, Y., et al. (2024). WildChat: 1M ChatGPT Interaction Logs in the Wild. arXiv preprint arXiv:2405.01470. https://arxiv.org/abs/2405.01470

[7] Ren, P., Chen, Z., Ren, Z., Wei, F., Ma, J., & de Rijke, M. (2021). Conversational Information Retrieval: A Survey. arXiv preprint arXiv:2003.04381. https://arxiv.org/abs/2003.04381

[8] Mehrotra, R., & Yilmaz, E. (2017). Task-Based Evaluation of Web Search Sessions. Proceedings of SIGIR 2017. DOI: 10.1145/3077136.3080797

[9] Mehrotra, R., et al. (2020). User Intent Modeling in Local Search. WWW Companion 2020. DOI: 10.1145/3366424.3383536

[10] Broder, A. (2002). A Taxonomy of Web Search. SIGIR Forum, 36(2). DOI: 10.1145/792550.792552

[11] Rose, D. E., & Levinson, D. (2004). Understanding User Goals in Web Search. Proceedings of WWW 2004. DOI: 10.1145/988672.988707

[12] Li, J., et al. (2024). Attribution, Citation, and Quotation: A Survey of Evidence-Based Generation in Large Language Models. arXiv preprint arXiv:2408.15396. https://arxiv.org/abs/2408.15396

[13] Zhang, H., et al. (2024). AGREE: Adaptation for Grounding Enhancement in Large Language Models. Proceedings of NAACL 2024. https://aclanthology.org/2024.naacl-long.346.pdf

[14] Surden, H. (2020). Artificial Intelligence and Law: An Overview. Georgia State University Law Review, 36(4). DOI: 10.2139/ssrn.3332487

[15] Liu, H., & Athey, S. (2021). The Impact of Artificial Intelligence on Legal Services. Annual Review of Law and Social Science, 17. DOI: 10.1146/annurev-lawsocsci-012620-093610

[16] American Bar Association. (2024). Formal Opinion 512: Generative Artificial Intelligence Tools. ABA Standing Committee on Ethics and Professional Responsibility. https://www.americanbar.org/news/abanews/aba-news-archives/2024/07/aba-issues-first-ethics-guidance-ai-tools/

[17] Yadav, P., et al. (2025). Retrieval-Augmented Generation (RAG) in Healthcare: A Comprehensive Review. Applied Sciences, 6(9), 226. https://www.mdpi.com/2673-2688/6/9/226

[18] Cohen, J. P., et al. (2024). On the Safety of Large Language Models for Clinical Use. npj Digital Medicine, 7. DOI: 10.1038/s41746-024-00890-9

[19] Liu, X., et al. (2023). Trustworthy AI in Healthcare: A Survey. ACM Computing Surveys, 55(8). DOI: 10.1145/3517341

[20] Yin, H., et al. (2021). Revisiting Spatial Relevance Models for Local Search. Proceedings of SIGIR 2021. DOI: 10.1145/3404835.3462894

[21] Noy, N., & Jain, A. (2020). Schema.org: Evolution of Structured Data on the Web. Communications of the ACM, 63(7), 72-79. DOI: 10.1145/3387107

[22] Zheng, L., et al. (2021). Schema-Guided Search and Recommendation with Structured Data. Proceedings of CIKM 2021. DOI: 10.1145/3459637.3482261

[23] Wang, S., et al. (2022). Understanding the Role of Reviews in Local Search. Information Processing & Management, 59(5). DOI: 10.1016/j.ipm.2022.102993

[24] Zheng, Z., et al. (2020). A Location-Aware Neural News Recommendation Model. Proceedings of CIKM 2020. DOI: 10.1145/3340531.3412040

[25] Weidinger, L., et al. (2022). Taxonomy of Risks Posed by Language Models. Proceedings of FAccT 2022. DOI: 10.1145/3531146.3533088

[26] Rashkin, H., et al. (2021). Measuring Attribution in Natural Language Generation Models. Proceedings of NeurIPS 2021. DOI: 10.48550/arXiv.2112.00861

[27] Jansen, B. J., Booth, D. L., & Spink, A. (2008). Determining the Informational, Navigational, and Transactional Intent of Web Queries. Information Processing & Management, 44(3), 1251-1266. DOI: 10.1016/j.ipm.2007.07.015

[28] Dalton, J., Xiong, C., & Callan, J. (2020). TREC CAsT 2019: The Conversational Assistance Track Overview. TREC 2019 Proceedings. arXiv:2003.13624. https://arxiv.org/abs/2003.13624

[29] Bendersky, M., Croft, W. B., & Smith, D. A. (2010). Analysis of Long Queries in a Large Scale Search Log. Proceedings of WSDM 2010. http://bendersky.github.io/pubs/2009-1.pdf

[30] Jones, C. B., & Purves, R. S. (2018). Progress and Challenges in Spatial Search of Text. Foundations and Trends in Information Retrieval, 12(2-3), 163-330. https://users.cs.cf.ac.uk/C.B.Jones/articles/FnTR-GIR2018.pdf

[31] Jones, R., & Klinkner, K. L. (2008). Beyond the Session Timeout: Automatic Hierarchical Segmentation of Search Topics in Query Logs. Proceedings of CIKM 2008. DOI: 10.1145/1458082.1458106

[32] Liu, T.-Y. (2009). Learning to Rank for Information Retrieval. Foundations and Trends in Information Retrieval, 3(3), 225-331. DOI: 10.1561/1500000016

[33] Robertson, S. E., & Walker, S. (1994). Some Simple Effective Approximations to the 2-Poisson Model for Probabilistic Weighted Retrieval. Proceedings of SIGIR 1994. DOI: 10.1007/978-1-4471-2099-5_24

[34] Fogg, B. J., Soohoo, C., Danielson, D. R., Marable, L., Stanford, J., & Tauber, E. R. (2003). How Do Users Evaluate the Credibility of Web Sites? A Study with Over 2,500 Participants. Proceedings of CHI 2003. DOI: 10.1145/642611.642632

[35] Zheng, L., Noroozi, V., & Yu, P. S. (2017). Joint Deep Modeling of Users and Items Using Reviews for Recommendation. Proceedings of WSDM 2017. DOI: 10.1145/3018661.3018665

[36] Quraishi, A., et al. (2024). Machine Learning Interpretability: Explaining AI Models to Humans. Xoffencerpublication.

[37] Quraishi, A., Soni, M., et al. (2024). Hybrid Disease Prediction Approach Leveraging Digital Twin and Metaverse Technologies for Remote Patient Monitoring. BMC Medical Informatics and Decision Making. DOI: 10.1186/s12911-024-02589-3

[38] Quraishi, A., et al. (2025). Dynamic Weight Allocation-Based Network Security and Anomaly Detection for Intelligent VANETs. Transactions on Emerging Telecommunications Technologies.

[39] Quraishi, A., et al. (2025). Edge AI-Based Dynamic Treatment Strategy Generation Model. IEEE Transactions on Consumer Electronics.

[40] Quraishi, A., et al. (2025). Post Quantum-Resistant Blind Signature Scheme for Consumer IoT Security. IEEE Transactions on Consumer Electronics.

APPENDIX A: DATA AVAILABILITY

The WildChat dataset is publicly available at: https://huggingface.co/datasets/allenai/WildChat-1M

Dataset version used: April 2024 release (14 parquet files, ~838,000 conversations)

APPENDIX B: AUTHOR CONTRIBUTIONS

Aadam Quraishi: Conceptualization, methodology development (VQMCS, LBAS), theoretical framework, statistical analysis, validation, supervision, writing—original draft, writing—review & editing.

Fatimah Quraishi: Data extraction, classification processing, visualization, writing—review & editing.

APPENDIX C: VQMCS IMPLEMENTATION DETAILS

This appendix provides implementation details sufficient for replication of the Vertical Query Mining and Classification System.

C.1 Keyword Lists by Vertical (Representative Samples)

Legal (87 patterns):

- High-weight (w=1.0): divorce attorney, personal injury lawyer, workers compensation claim, child custody lawyer, criminal defense attorney, immigration lawyer, estate planning attorney
- Medium-weight (w=0.6): sue for, legal advice about, can I file, statute of limitations, liability for
- Low-weight (w=0.3): is it legal, legal question, rights regarding

Medical (94 patterns):

- High-weight (w=1.0): doctor near me, specialist for, treatment for [condition], symptoms of [disease], cardiologist, dermatologist, orthopedic surgeon
- Medium-weight (w=0.6): should I see a doctor, medical advice, prescription for

- Low-weight (w=0.3): is [symptom] serious, health question

HVAC (42 patterns):

- High-weight (w=1.0): HVAC repair, AC not working, furnace replacement, heating contractor, air conditioning service
- Medium-weight (w=0.6): AC makes noise, heater won't turn on
- Low-weight (w=0.3): house too hot, temperature problems

Automotive (56 patterns):

- High-weight (w=1.0): car dealership, mechanic near me, auto repair shop, buy used car, transmission repair
- Medium-weight (w=0.6): car making noise, check engine light
- Low-weight (w=0.3): car problems, vehicle issues

Roofing (38 patterns):

- High-weight (w=1.0): roofing contractor, roof repair, roof replacement, storm damage roof, shingle replacement
- Medium-weight (w=0.6): roof leaking, missing shingles
- Low-weight (w=0.3): roof problems, ceiling water stain

Home Services (31 patterns):

- High-weight (w=1.0): plumber near me, electrician, handyman service, pest control, cleaning service
- Medium-weight (w=0.6): toilet won't flush, power outlet not working
- Low-weight (w=0.3): home repair, fix my house

C.2 Scoring Parameters

Parameter	Value	Description
α (context weight)	0.7	Weight for contextual signals from adjacent turns
High-weight terms	$w = 1.0$	Explicit service + provider patterns
Medium-weight terms	$w = 0.6$	Implicit service need patterns
Low-weight terms	$w = 0.3$	General domain vocabulary

Vertical Thresholds (T_v):

Vertical	Threshold	Rationale
Legal	2.5	Higher threshold due to frequent legal terminology in non-service contexts
Medical	2.1	Moderate threshold; health queries common but often informational
HVAC	1.8	Lower threshold; HVAC terms rarely appear outside service contexts
Automotive	1.9	Moderate; automotive terms appear in hobbyist/news contexts
Roofing	1.7	Lower threshold; roofing terms highly service-specific
Home Services	1.6	Lowest threshold; diverse category requires sensitivity

C.3 Pseudocode

```
FUNCTION VQMCS_Classify(conversation):
  scores = {}
  FOR each vertical v IN [Legal, Medical, HVAC, Auto, Roofing, Home]:
    scores[v] = 0
    FOR each message m IN conversation:
      FOR each pattern p IN keyword_list[v]:
        IF regex_match(p, m.text):
          scores[v] += weight[p] * count_matches(p, m.text)
      # Add contextual signal from adjacent turns
      scores[v] += α * context_signal(m, v)

  # Assign to highest-scoring vertical above threshold
  best_v = argmax(scores)
  IF scores[best_v] >= threshold[best_v]:
    RETURN best_v
  ELSE:
    RETURN None  # Not a local service query
```

C.4 Processing Environment

- Platform: Google Colaboratory (Python 3.10)
- Libraries: pandas 2.0.3, pyarrow 12.0.1, regex 2023.6.3
- Hardware: Standard Colab runtime (12GB RAM)
- Throughput: ~8,000 conversations/minute

- Total processing time: 70-140 minutes for full dataset

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